C106: An Exact Coupling Witness in a Variational Two-Site Hénon Lattice

Route-A exploratory research note

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Abstract

We record an exact, deliberately small audit of a two-site variational Hénon map. The map is reversible and symplectic by construction. At rational parameters we certify two fixed points and one genuine synchronous period-two orbit using rational arithmetic, and compare its finite monodromy polynomial with the uncoupled control. Coupling changes both the trace and the quadratic coefficient. This is an A1 low-period witness only: no completeness theorem, transfer-operator owner, or Fredholm determinant is claimed.

1 Model and scope

Let $q = (x, y)$ and $p = (u, v)$, and define

$$U(q) = \frac{a}{2}(x^2 + y^2) - \frac{x^3 + y^3}{3} - \frac{\kappa}{2}(x - y)^2, \qquad F(q, p) = (\nabla U(q) - p, q), \tag{1}$$

with $(a, \kappa) = (7, 1/4)$. In the coordinate order (q_1, q_2, p_1, p_2) use $\Omega = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$ and $R(q, p) = (p, q)$. The exact producer and independent checker are included in the accompanying artifact.

For every twice differentiable U,

$$DF = \begin{pmatrix} D^2U & -I \\ I & 0 \end{pmatrix}, \qquad DF^T \Omega DF = \Omega, \qquad \det DF = 1,$$

and direct substitution gives $RFR = F^{-1}$. These identities are checked over $\mathbb{Q}$ at all reported orbit points and at three independent rational samples.

2 Certified low-period data

The fixed points $(0, 0; 0, 0)$ and $(5, 5; 5, 5)$ satisfy $\nabla U(q) = 2q$. A non-fixed period-two orbit is

$$z_0 = (3, 3; 6, 6), \qquad z_1 = (6, 6; 3, 3), \qquad F(z_0) = z_1, \quad F(z_1) = z_0.$$

For $M = DF(z_1)DF(z_0)$, exact arithmetic gives the following table.

object	period	$\operatorname{tr} M$	$\det(I - zM)$
origin	1	$27/2$	$1 - \frac{27}{2}z + \frac{95}{2}z^2 - \frac{27}{2}z^3 + z^4$
synchronous fixed	1	$-13/2$	$1 + \frac{13}{2}z + \frac{25}{2}z^2 + \frac{13}{2}z^3 + z^4$
synchronous 2-cycle	2	$-47/4$	$1 + \frac{47}{4}z + \frac{141}{4}z^2 + \frac{47}{4}z^3 + z^4$

The two-cycle has determinant one. With the same a and the same synchronous states but $\kappa = 0$, the last polynomial becomes

$$1 + 14z + 51z^2 + 14z^3 + z^4.$$

Thus the coupled-minus-uncoupled trace difference is $9/4$, while the z^2 coefficient difference is $-63/4$. This is a finite-dimensional product-versus-coupled control; it is not a Fredholm determinant.

3 Route-A boundary

The ledger certifies only the displayed low-period witnesses. It does not establish a complete primitive-orbit enumeration, a Markov partition, a common anisotropic function space, nuclearity, a trace formula, or a zero-counting theorem. Accordingly the package records

$$(A1, A2, A3, A4) = \big(\text{partial low-period},\ \text{operator owner open},\ \text{not addressed},\ \text{fail}\big),$$

with overall status `ROUTE_A_EXPLORATORY`. In the formal evaluator vocabulary this is `A1_WEAK` and `A2_FAIL`, with the qualification that the displayed finite operator owner remains open. In particular, no arithmetic local data, Euler factors, root numbers, automorphy, or Hilbert–Pólya operator is asserted.

4 Reproducibility

The JSON evidence is canonicalized and checked byte-for-byte. A separate checker reconstructs the gradients, Jacobians, symplectic form, reversor, monodromy, and uncoupled control. A SymPy script supplies an independent symbolic check; replay and eleven hostile semantic mutations complete the integrity audit. All calculations use exact rationals and no random input.

Data availability. The scripts and evidence are in the C106 package directory.
The scope firewall is `NO_BAD_EULER_OR_ROOT_NUMBER`.